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Multiphysics applications across scales: integration of gravity, magnetic and electromagnetic data for solving exploration challenges.

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Summary

As geophysicists we have at our disposal a range of approaches to probing the sub-surface. We can measure a range of physical properties at a range of scales. By carefully combining disparate data in multiphysics workflows, the strengths in one method can compensate for the weaknesses in another. This leads to a more robust characterization of earth structure and properties overall. Multiphysics workflows can be tailored to address a range of challenges from regional mapping, through prospect evaluation, to high resolution mapping of the shallow sub-surface.

In this paper we present exploration examples at each of these scales and discuss the benefits that a multiphysics approach brings to each.

Introduction

Multiphysics approaches combine different geophysical data types, which measure different properties of the earth, in workflows to provide a better or more complete understanding of the sub-surface than can be achieved by any method taken by itself. Such workflows are varied and can be applied to sub-surface characterization challenges across a range of scales (Figure 1).

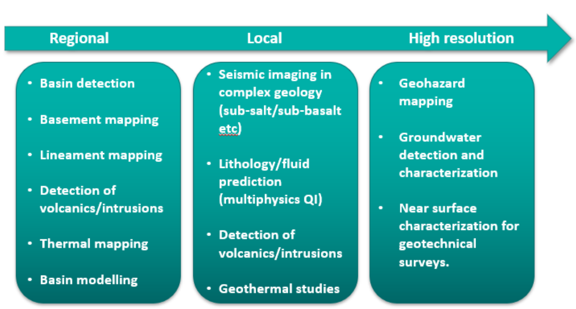

Figure 1: Multiphysics workflows that combine gravity, magnetic, electromagnetic and seismic data can be applied to sub-surface characterization challenges across a range of scales.

At a regional scale, gravity and magnetic data can be combined to locate basins, understand basement depth, lineaments and detect volcanic intrusions. Combined with large scale electromagnetic and seismic data it is possible to glean insights into thermal history and crustal architecture (Meju et al., 2023)

More locally, electromagnetic methods have been used to improve seismic imaging in geologically complex areas (for example around salt and basalt) and as a complement to seismic reservoir characterization to reduce uncertainty in prospect ranking and appraisal. Applications to 4D reservoir monitoring are being contemplated, both during hydrocarbon extraction and as a tool to monitor CO2 injection into sub-surface storage sites (e.g. Colombo et al., 2018; Alvarez et al., 2017, 2018; Subagjo et al. 2018).

At a higher resolution still, multiphysics workflows are used in the characterization of water resources, seafloor minerals, hazard mapping and near surface characterization for geotechnical studies among other applications (e.g. Attias et al, 2020; Kowalczyk, 2008).

Regional scale.

Figure 2 shows an example of the interpretation of gravity and magnetic data to identify basins and identify areas likely to contain shallow intrusives and/or shallow crystalline basement on a regional scale (Figure 2).

However one of the challenges to be addressed is to characterize more completely the depth to and nature of the 'basement' (which can have several different definitions), and distinguish volcanics from potentially prospective carbonate build ups at depth. This can be accomplished by comparing 'basement' depth picked from seismic (typically the deepest strong reflector), with that derived from magnetotelluric data (usually defined as a relatively sharp increase in resistivity that is associated with a transition from sedimentary to metamorphic or volcanic rock types) and magnetic data (the depth to magnetic basement).

Figure 3 shows the difference in depth between a seismic pick defining the top of a potentially prospective section, and the magnetic basement as derived from ship borne magnetic data. In area A, there is a positive difference between the (shallower) seismic pick and the depth to magnetic basement. There is an increase in resistivity derived from magnetotelluric and controlled source electromagnetic (CSEM) data coincident with the seismic pick consistent with a transition to carbonates or more compacted sediment,

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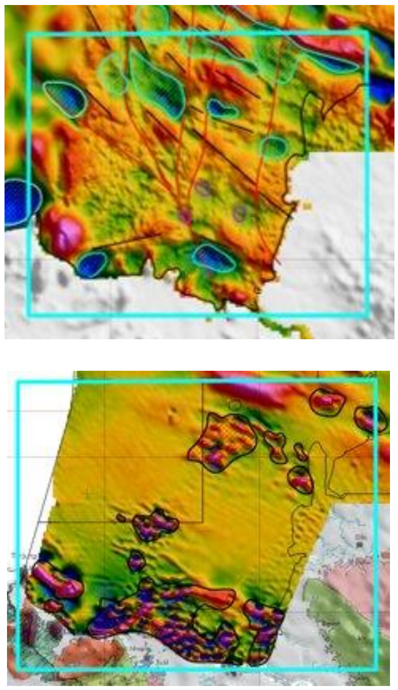

Figure 2: Top: potential basins mapped on the Borneo margin in gravity data (complete Bouger anomaly). Bottom: Shallow intrusives and/or shallow basement mapped in magnetic data in the same area.

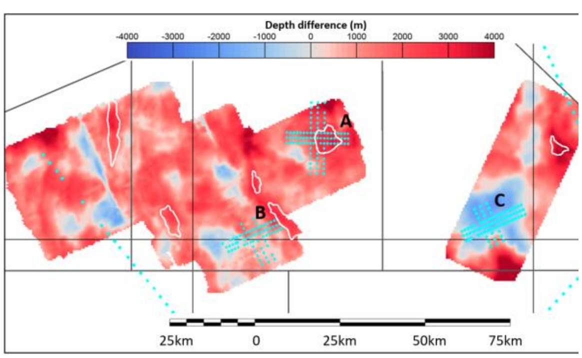

Figure 3: Difference between the seismic pick of a potentially prospective section, and the magnetic basement. Comparing these with resistivity sections derived from MT an CSEM data highlights inconsistencies in the picks that are indicative of lithological variations. Areas A, B and C are discussed in the text. Turquoise symbols show the position of CSEM/MT receivers.

but lower than would be expected for crystalline basement. Taken together these observations suggest that in this area the seismic pick represents the top of a sedimentary section as hoped.

In contrast in area B, the magnetic depth to basement and the seismic pick are almost coincident, suggesting that here the seismic pick is in fact crystalline basement. Based on the resistivity profile, the sedimentary horizon sought lies shallower in the section, prompting a re-evaluation of the seismic data.

Area C is the most intriguing: Here the seismic pick lies considerably deeper than the magnetic basement. It is not clear geologically what this represents, but again a re-evaluation of the seismic data is order to understand this more fully.

Local scale

Multiphysics combinations of gravity, magnetic, CSEM and MT are used at a more local scale for prospect characterization. A first example of this again comes from offshore Borneo, where the challenge was to determine whether a seismically identified prospect was a prospective carbonate build up, or more likely to be a volcanic feature (figure 4).

Gravity and magnetic data were inverted for 3D distributions of density and susceptibility respectively constrained by the seismic structure, and these compared to the seismic velocity. The prospect was found to coincide with a zone of high density, high susceptibility and high velocity making the interpretation of the feature as being predominantly volcanic more likely and significantly increasing the risk associated with the prospect.

The use of CSEM in multiphysics workflows alongside more traditional seismic quantitative interpretation workflows is becoming increasingly common. Approaches range from multiphysics attribute analysis applications (for example Alvarez et al. 2018; Saleh et al. 2023) to petrophysical joint inversion approaches (for example Alvarez et al, 2017; Andreis et al, 2018) This is a particularly powerful approach when there is an uncertainty in the hydrocarbon saturation (for example between commercial and sub-commercial saturations) that cannot be resolved on the basis of seismic alone.

An area where this problem can be particularly acute is in carbonate reservoirs, where there can be little or no sensitivity to fluid type and/or saturation in seismic data. Carbonate reservoirs have traditionally been thought to be challenging for CSEM methods, because of the potentially high reservoir resistivity when wet. MacGregor (2012) demonstrates and early example of an integrated interpretation approach in a chalk reservoir that successfully

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distinguished between tight, porous wet and porous charged chalk. More recent well log based studies also show that combining seismic and CSEM methods is a promising approach in Miocene carbonates offshore Borneo.

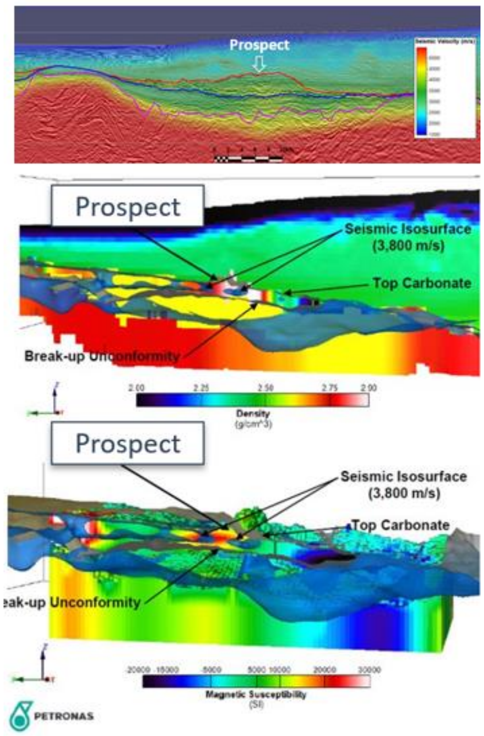

Figure 4: Top: A seismically defined prospect which could be either a carbonate build up or a volcanic feature was mapped. Seismic velocity is overlain. Middle: density from 3D inversion of gravity data. Bottom: Susceptibility from 3D inversion of magnetic data. In both cases the inversions were constrained by seismic structure. Integrated interpretation suggests the volcanic interpretation of the feature is more likely.

The top panel of Figure 5 shows a multiphysics cross plot combining acoustic impedance with the resistivity. The water wet carbonates in this area overlie a reservoir interval of stacked carbonate and clastic reservoirs. Using seismic alone it is hard to characterize these reservoirs (and indeed the overlying carbonate in places makes even imaging them a challenge). There is overlap in resistivity between the charged reservoirs and the overlying carbonate, although on average the charged reservoirs have a higher resistivity than the water wet carbonates. However, seismic and CSEM derived attributes taken together can distinguish between these lithology and fluid scenarios.

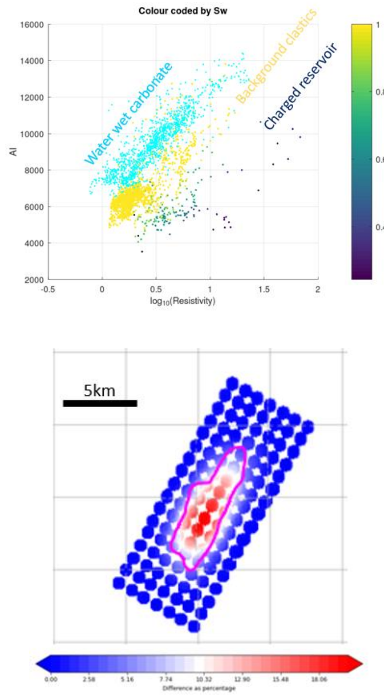

Figure 5: Top: Multiphysics crossplot of acoustic impedance and resistivity, highlighting the separation in this space of charged and wet reservoirs in the presence of water wet carbonates. Bottom: CSEM sensitivity for the case of a reservoir beneath the water wet carbonates. Common offset plot at 5km for a 0.2Hz signal is shown. See text for details.

The bottom panel of figure 5 shows the results of a CSEM sensitivity analysis for this scenario. The background model was constructed based on seismic velocity. A velocity-resistivity relationship (the Faust equation) was calibrated separately for clastic and carbonate intervals based on well log data. This was used to derive a background (water wet) resistivity volume from seismic velocity, which encompassed both clastics and carbonates, and crucially captured the variability in the carbonate section, which can be significant. The CSEM response of a charged sub-carbonate reservoir was then compared to the response of the water wet background (Figure 5 lower plot). Differences in excess of 15% are observed, well above the typical CSEM noise floor in modern surveys. It is hoped that this promising application will be tested in the field shortly.

High resolution

High resolution and near surface applications of multiphysics include ultra-high resolution AUV based CSEM and magnetic studies of seafloor minerals

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(Kowalczyk, 2008), hazard mapping and infrastructure inspection as well as onshore, transition zone and offshore CSEM mapping of fresh water resources (e.g. Attias et al., 2020).

CSEM integrated with seismic has been used to characterize the shallow sub-surface in a number of studies, particularly in the characterization of shallow gas and hydrates. Karpiah et al. 2023 present an approach to this challenge based on the joint inversion of CSEM data seismic traveltimes derived from standard seismic survey data. A cross gradient constraint between resistivity and velocity models is applied to ensure structural consistency. An example of the application of this approach is shown in figure 6. The left and right panels show the resistivity and velocity respectively at a depth of 25m below the seafloor, whilst the middle panel shows an interpretation of shallow and intermediate depth faults mapped from 2D seismic data. There is a clear correlation between these features and the recovered velocity and resistivity.

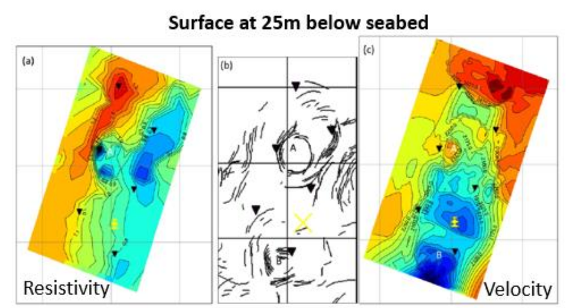

Figure 6: Resistivity and velocity at 25m below seafloor from the joint inversion of CSEM and seismic traveltime data (Karpiah et al., 2023). The middle panel shows an interpretation of shallow and intermediate depth faults from 2D seismic.

Conclusions

Multiphysics approaches range from semi quantitative attribute analysis and integration interpretation workflows, to quantitative joint inversion approaches, and can be applied across a range of scales and applications. Key to their success is the identification of the combination of data types and attributes that address an uncertainty or ambiguity in sub-surface characterization.

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